Lecture 2: Linear dimensionality reduction

TTIC 31220: Unsupervised Learning and Data Analysis

Instructor: Karen Livescu

TTI-Chicago

March 25, 2026

Announcements

Consent requests etc.

- Class is oversubscribed for now; if you decide to drop the class, please update your registration on registrar site as soon as possible
- Consent requests: Submit online; mention your prereqs/equivalents + grades

Announcements

- Tutorial 1 was a video (sorry for the late notice!); please bring any questions to TA office hours after class today
- Tutorial 2 next Wednesday: Kernels
- Homework 1 & Quiz 1 tentative schedule:
  - HW1 released in the next few days
  - HW1 deadline April 9
  - Quiz 1 April 14

Collaboration policy

- You are encouraged to discuss homework and any course material with others
- Submitted material (writeups, code) must be produced on your own
- Don't submit anyone else's words (including an LLM's) as your own

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.
- But LLMs are a poor approximation of course staff. Suggested workflow: Use the materials we provide first, then talk to course staff and friends, then talk to an LLM, then double-check the LLM.

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.
- But LLMs are a poor approximation of course staff. Suggested workflow: Use the materials we provide first, then talk to course staff and friends, then talk to an LLM, then double-check the LLM.
- Also use LLMs (or any other tool) to proofread your writing for grammar/typos.

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.
- But LLMs are a poor approximation of course staff. Suggested workflow: Use the materials we provide first, then talk to course staff and friends, then talk to an LLM, then double-check the LLM.
- Also use LLMs (or any other tool) to proofread your writing for grammar/typos.
- Submitted solution must be your own words.

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.
- But LLMs are a poor approximation of course staff. Suggested workflow: Use the materials we provide first, then talk to course staff and friends, then talk to an LLM, then double-check the LLM.
- Also use LLMs (or any other tool) to proofread your writing for grammar/typos.
- Submitted solution must be your own words.
- If you used LLMs, describe how you did so and what you found most useful!

Review from last time: Supervised learning setup

- **Basic ingredients of supervised learning**

Review from last time: Supervised learning setup

- **Basic ingredients of supervised learning**
  - Training data: pairs $(\mathbf{x}_i, y_i)$ labeled by a "teacher"
  - Function (hypothesis) class F, usually parametric $\{f(\mathbf{x}; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
  - Loss function ℓ penalizing for predicting $\widehat{y}$ when ground truth is y

Review from last time: Supervised learning setup

- **Basic ingredients of supervised learning**
  - Training data: pairs $(\mathbf{x}_i, y_i)$ labeled by a "teacher"
  - Function (hypothesis) class F, usually parametric $\{f(\mathbf{x}; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
  - Loss function ℓ penalizing for predicting $\widehat{y}$ when ground truth is y
- **Empirical risk minimization**: learning = minimizing empirical loss

$$\boldsymbol{\theta}^* = \operatorname*{argmin}_{\boldsymbol{\theta}} \sum_i \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i)$$

Review from last time: Supervised learning setup

- **Basic ingredients of supervised learning**
  - Training data: pairs $(\mathbf{x}_i, y_i)$ labeled by a "teacher"
  - Function (hypothesis) class F, usually parametric $\{f(\mathbf{x}; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
  - Loss function ℓ penalizing for predicting $\widehat{y}$ when ground truth is y
- **Empirical risk minimization**: learning = minimizing empirical loss

$$\boldsymbol{\theta}^* = \operatorname*{argmin}_{\boldsymbol{\theta}} \sum_i \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i)$$

- **Note**: ℓ is usually related to the real task, but often is not exactly the same as the task loss (that we really care about)

Follow-up from last time: Learning hyperparameters

Regularized ERM: minimize

$$\sum_{i=1}^{N} \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i) + \lambda R(\boldsymbol{\theta})$$

- Discussed last time: Regularization term weight can not be learned from the model training data
- Question: Can we automatically set λ?
- Short answer: Yes, there are a number of methods
  - Tuning hyperparameters on held-out data: "Manually" or automatically (e.g., Bayesian hyperparameter optimization, gradient descent on held-out set)
  - Note: Hyperparameter learning algorithms also have hyperparameters!

Unsupervised learning setup

- Data set $X = \{\mathbf{x}_i\}$ (no labels)
- Training set, validation set, test set just like in supervised settings
- An objective J defined in terms of $\mathbf{x}_i$
  typically parametric $\{J(X; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
- Learning = minimizing the objective over the training data

$$\min_{\boldsymbol{\theta}} \{J(X; \boldsymbol{\theta}) + \lambda R(\boldsymbol{\theta})\}$$

  Note: J may decompose over examples $\mathbf{x}_i$ or not
- As in supervised learning, may overfit with more complex models*, so need regularization and model selection (validation set)
- (* Recent large neural models seem to tolerate more complexity, but regularization is still useful)
- What is J? That is the big question

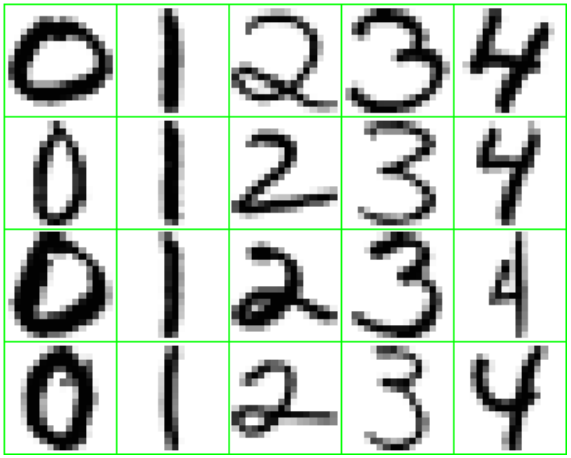
Representation learning: Linear dimensionality reduction

- Principal component analysis (PCA)
- Canonical correlation analysis (CCA)

Dimensionality reduction

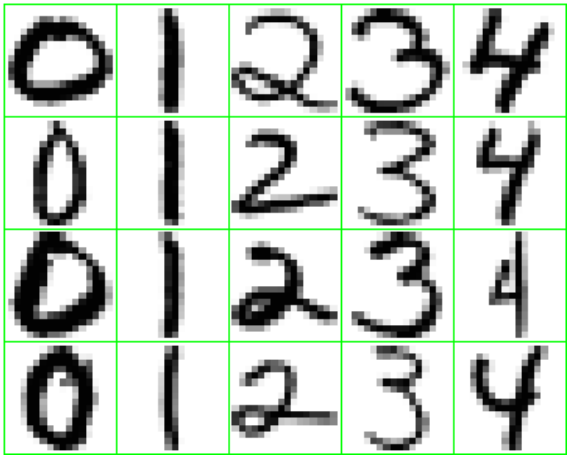
Dimensionality reduction

- Consider data drawn from a high-dimensional space
- Example: handwritten digits $\in \mathbb{R}^{256}$ (16×16=256 pixels)



Dimensionality reduction

- Consider data drawn from a high-dimensional space
- Example: handwritten digits $\in \mathbb{R}^{256}$ (16×16=256 pixels)

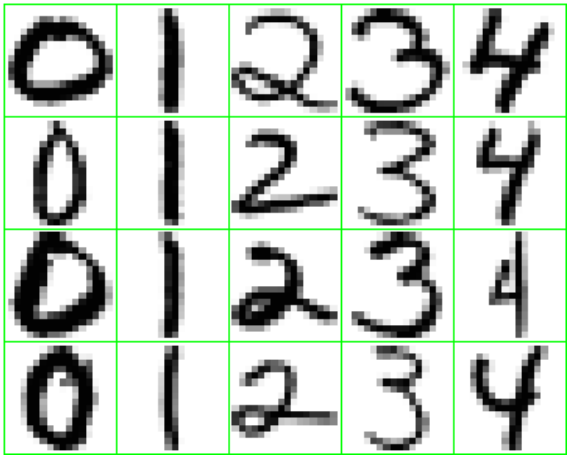


- Intuition: we can explain most of the variations in these digits with only a few dimensions (How do we know? And why would we want to?). So, we seek a projection

$$P(\mathbf{x}; \boldsymbol{\theta}) : \mathbb{R}^d \rightarrow \mathbb{R}^k, \qquad k \ll d$$

Dimensionality reduction

- Consider data drawn from a high-dimensional space
- Example: handwritten digits $\in \mathbb{R}^{256}$ (16×16=256 pixels)



- Intuition: we can explain most of the variations in these digits with only a few dimensions (How do we know? And why would we want to?). So, we seek a projection

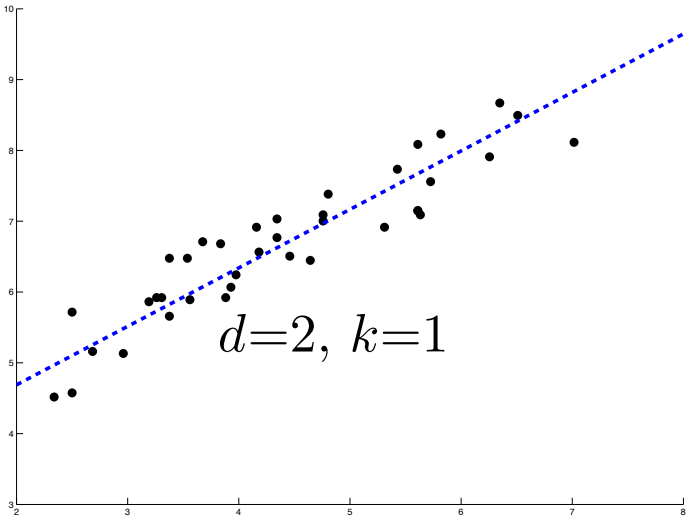
$$P(\mathbf{x}; \boldsymbol{\theta}) : \mathbb{R}^d \rightarrow \mathbb{R}^k, \qquad k \ll d$$

- We fit P by

$$\min_{\boldsymbol{\theta}} J\left(\mathbf{x}_1, \ldots, \mathbf{x}_N, P(\mathbf{x}_1), \ldots, P(\mathbf{x}_N)\right)$$

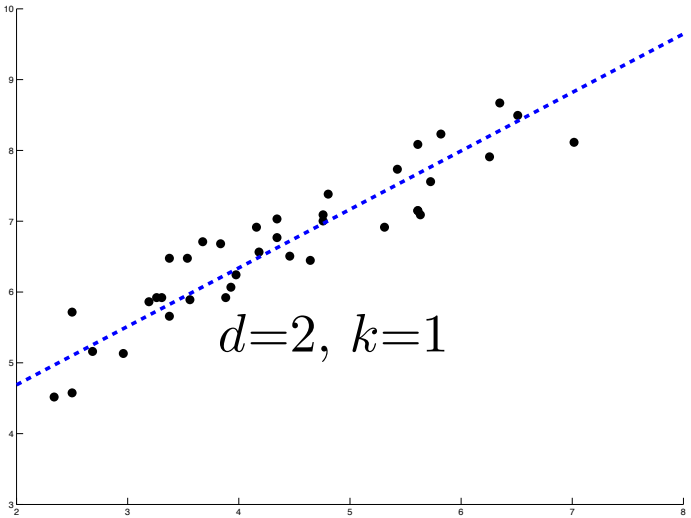
Dimensionality reduction

- d: the "nominal" or "ambient" or "input" dimensionality
- There is often a lower *intrinsic* dimensionality: a property of the generating process $\Rightarrow$ the data lie on (or near) a subspace of dimensionality $k < d$.



Dimensionality reduction

- d: the "nominal" or "ambient" or "input" dimensionality
- There is often a lower *intrinsic* dimensionality: a property of the generating process $\Rightarrow$ the data lie on (or near) a subspace of dimensionality $k < d$.



Dimensionality reduction seeks to recover simple structure that "explains" the data

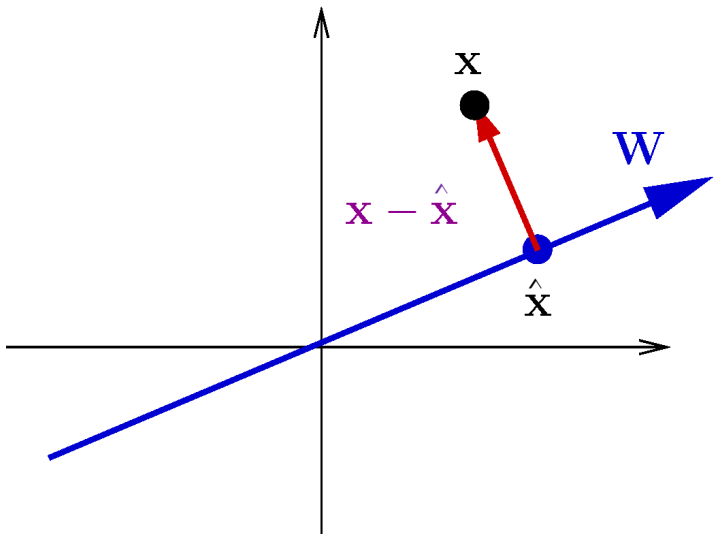
- by finding the underlying subspace
- equivalently, by recovering hidden variables (directions of variation)

Linear dimensionality reduction

- Assume for now that the data points have zero mean
- Linear subspace is defined by a $d \times k$ matrix $\mathbf{W}$ containing k direction vectors
- Approximation (reconstruction) of $\mathbf{x}$ as a weighted combination of the vectors in $\mathbf{W}$:

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{z}$$

- $\mathbf{z}$ is a data point-specific vector of k weights
- $\mathbf{z}$ can be seen as a low-dimensional representation of $\mathbf{x}$



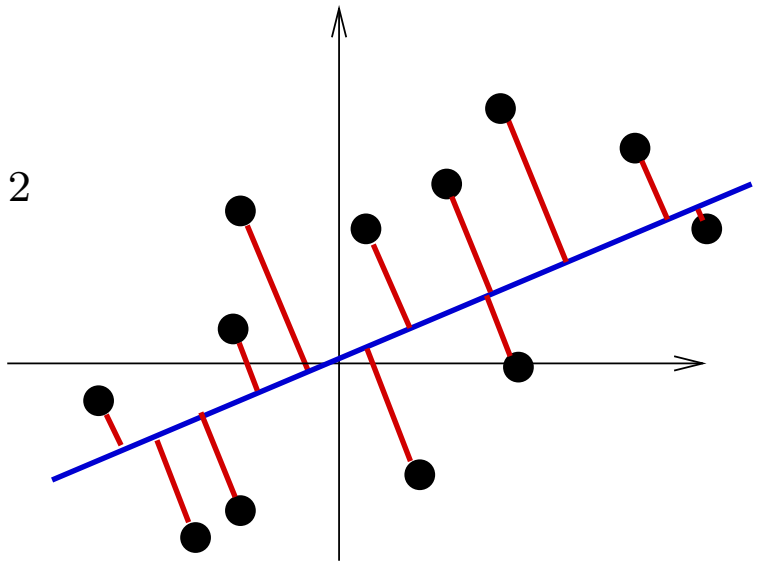
Principal component(s) analysis (PCA)

Proposed by Pearson (1901) and further (independently?) developed by Hotelling in 1933

- A linear dimensionality reduction technique
- **Objective**: minimize the sum of *residuals* subject to orthogonality of projection directions (= "principal directions")

$$\mathbf{W}^* = \underset{\mathbf{W}}{\operatorname{argmin}} \sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$$

$$\text{s.t. } \mathbf{W}^T\mathbf{W} = I$$



Principal component(s) analysis (PCA)

PCA is the solution to the minimum-residual projection problem, which is as follows:

- Let the columns of $\mathbf{X}$ be the data points ($d \times N$ matrix)
- Construct the $d \times d$ *data covariance matrix* $\mathbf{S} = \frac{1}{N}\mathbf{X}\mathbf{X}^T$;
- Let $\phi_1, \ldots, \phi_d$ be the orthonormal eigenvectors of $\mathbf{S}$ corresponding to the eigenvalues $\lambda_1 \geq \ldots \geq \lambda_d$.

$$\phi_l^T \phi_j = \begin{cases} 0 & \text{if } l \neq j, \\ 1 & \text{for } l = j. \end{cases}$$

- Then the optimal k-dim linear subspace turns out to be given by

$$\mathbf{W} = \mathbf{\Phi} = [\phi_1, \ldots, \phi_k].$$

PCA: maximum variance subspace formulation

- An alternative criterion: maximize the *variance* of the projection

$$\begin{aligned} \mathbf{W}^* &= \underset{\mathbf{W}}{\operatorname{argmax}} \operatorname{tr}\left(\mathbf{W}^T \mathbf{X}\mathbf{X}^T \mathbf{W}\right) \\ \text{s.t. } \mathbf{W}^T\mathbf{W} &= I \end{aligned}$$

- Leads to the same solution as the minimum-residual formulation
- We next prove this claim (partially)...

(Partial) derivation and proof of equivalence

For the $k = 1$ case:

- $\mathbf{w}_j \in \mathbb{R}^d$: j^{th} principal direction
- $\mathbf{x}_i \in \mathbb{R}^d$: i^{th} data point
- $\mathbf{z}_i \in \mathbb{R}^k$: the corresponding projection
- $\tilde{\mathbf{z}}_j \in \mathbb{R}^N$: j^{th} component of all of the projections

(Partial) derivation and proof of equivalence

For the $k = 1$ case:

- $\mathbf{w}_j \in \mathbb{R}^d$: j^{th} principal direction
- $\mathbf{x}_i \in \mathbb{R}^d$: i^{th} data point
- $\mathbf{z}_i \in \mathbb{R}^k$: the corresponding projection
- $\tilde{\mathbf{z}}_j \in \mathbb{R}^N$: j^{th} component of all of the projections

$$
\begin{aligned}
J(\mathbf{w}_1, \tilde{\mathbf{z}}_1) &= \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i - z_{i1}\mathbf{w}_1\|^2 = \frac{1}{N}\sum_{i=1}^{N} (\mathbf{x}_i - z_{i1}\mathbf{w}_1)^T(\mathbf{x}_i - z_{i1}\mathbf{w}_1) \\
&= \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i^T\mathbf{x}_i - 2z_{i1}\mathbf{w}_1^T\mathbf{x}_i + z_{i1}^2\mathbf{w}_1^T\mathbf{w}_1 \\
(\mathbf{w}_1^T\mathbf{w}_1 = 1) &= \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i^T\mathbf{x}_i - 2z_{i1}\mathbf{w}_1^T\mathbf{x}_i + z_{i1}^2
\end{aligned}
$$

(Partial) derivation and proof of equivalence

Taking derivatives of $J()$ w.r.t. z_{i1}:

$$\frac{\partial J(\mathbf{w}_1, \tilde{\mathbf{z}}_1)}{\partial z_{i1}} = \frac{1}{N}\left[-2\mathbf{w}_1^T\mathbf{x}_i + 2z_{i1}\right] = 0 \Longrightarrow z_{i1} = \mathbf{w}_1^T\mathbf{x}_i$$

This shows that the optimal reconstruction coefficients (representation) z_{i1} are the orthogonal projection of the data onto the first principal direction $\mathbf{w}_1$.

(Partial) derivation and proof of equivalence

Taking derivatives of $J()$ w.r.t. z_{i1}:

$$\frac{\partial J(\mathbf{w}_1, \tilde{\mathbf{z}}_1)}{\partial z_{i1}} = \frac{1}{N}\left[-2\mathbf{w}_1^T\mathbf{x}_i + 2z_{i1}\right] = 0 \Longrightarrow z_{i1} = \mathbf{w}_1^T\mathbf{x}_i$$

This shows that the optimal reconstruction coefficients (representation) z_{i1} are the orthogonal projection of the data onto the first principal direction $\mathbf{w}_1$. Plugging back in:

(Partial) derivation and proof of equivalence

Taking derivatives of $J()$ w.r.t. z_{i1}:

$$\frac{\partial J(\mathbf{w}_1, \tilde{\mathbf{z}}_1)}{\partial z_{i1}} = \frac{1}{N}\left[-2\mathbf{w}_1^T\mathbf{x}_i + 2z_{i1}\right] = 0 \Longrightarrow z_{i1} = \mathbf{w}_1^T\mathbf{x}_i$$

This shows that the optimal reconstruction coefficients (representation) z_{i1} are the orthogonal projection of the data onto the first principal direction $\mathbf{w}_1$. Plugging back in:

$$J(\mathbf{w}_1) = \frac{1}{N}\sum_{i=1}^{N}[\mathbf{x}_i^T\mathbf{x}_i - z_{i1}^2] = \text{const} - \frac{1}{N}\sum_{i=1}^{N} z_{i1}^2$$

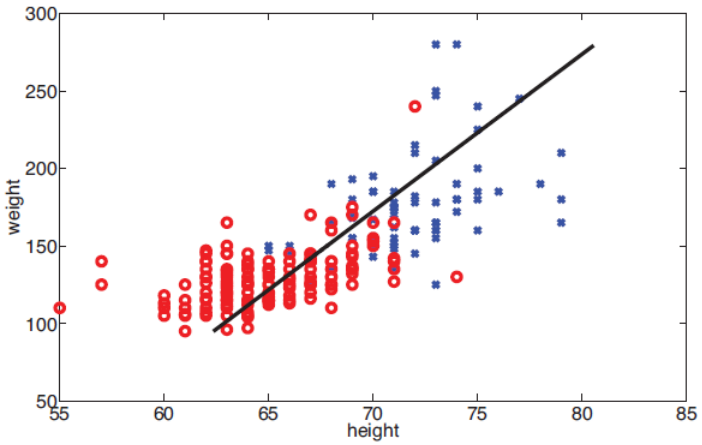
Now let's compute the variance of the projection:

$$\text{var}\,\tilde{\mathbf{z}}_1 = \mathbb{E}\left[\tilde{\mathbf{z}}_1^2\right] - \left(\mathbb{E}\left[\tilde{\mathbf{z}}_1\right]^2\right) = \frac{1}{N}\sum_{i=1}^{N} z_{i1}^2 - 0 \text{ (since } \mathbb{E}[\mathbf{x}] = 0 \Longrightarrow \mathbb{E}[\tilde{\mathbf{z}}_1] = 0)$$

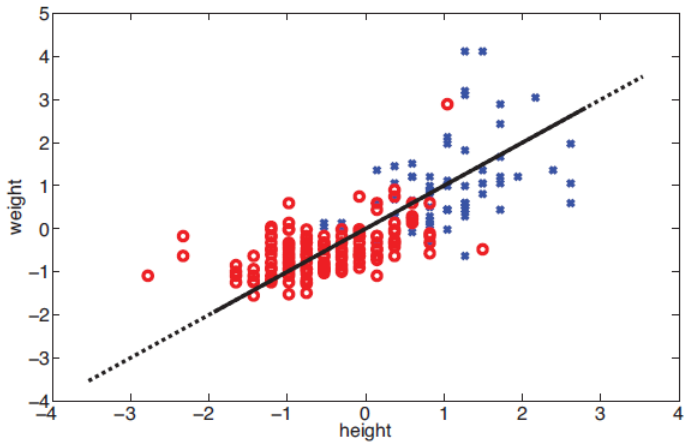
So, $J(\mathbf{w}_1) = \text{const} - \text{var}[\tilde{\mathbf{z}}_1]$, and minimizing $J()$ is equivalent to maximizing the variance of the projected data!

Negative implications of variance maximization

PCA can be fooled by the scale of the input dimenions:



(a)



(b)

So it is common practice to standardize the data first to unit variance in each (original) dimension

(Partial) derivation and proof of equivalence

We now solve for $\mathbf{w}_1$ by maximizing the variance of the projection, subject to $\|\mathbf{w}_1\| = 1$. First we rewrite the variance:

$$\frac{1}{N}\sum_{i=1}^{N} z_{i1}^2 = \frac{1}{N}\sum_{i=1}^{N} \mathbf{w}_1^T \mathbf{x}_i \mathbf{x}_i^T \mathbf{w}_1 = \mathbf{w}_1^T \mathbf{S} \mathbf{w}_1$$

(Partial) derivation and proof of equivalence

We now solve for $\mathbf{w}_1$ by maximizing the variance of the projection, subject to $\|\mathbf{w}_1\| = 1$. First we rewrite the variance:

$$\frac{1}{N}\sum_{i=1}^{N} z_{i1}^2 = \frac{1}{N}\sum_{i=1}^{N} \mathbf{w}_1^T \mathbf{x}_i \mathbf{x}_i^T \mathbf{w}_1 = \mathbf{w}_1^T \mathbf{S} \mathbf{w}_1$$

Then we maximize this variance subject to the constraint (using a Largrange multiplier to enforce the constraint):

$$\tilde{J}(\mathbf{w}_1) = \mathbf{w}_1^T \mathbf{S} \mathbf{w}_1 + \lambda_1 \left(1 - \mathbf{w}_1^T \mathbf{w}_1\right)$$

Now taking the derivative and setting to 0:

$$\begin{aligned} \frac{\partial \tilde{J}(\mathbf{w}_1)}{\partial \mathbf{w}_1} &= 2\mathbf{S}\mathbf{w}_1 - 2\lambda_1 \mathbf{w}_1 = 0 \\ \implies \mathbf{S}\mathbf{w}_1 &= \lambda_1 \mathbf{w}_1 \end{aligned}$$

So the direction $\mathbf{w}_1$ that maximizes the variance is an eigenvector of $\mathbf{S}$! But which eigenvector?

(Partial) derivation and proof of equivalence

- Left multiplying the eigenproblem by $\mathbf{w}_1$, we find that $\mathbf{w}_1^T \mathbf{S} \mathbf{w}_1 = \lambda_1$
- That is, **the projected data variance is the eigenvalue** λ_1
- Since we want to maximize the variance, we want to pick the eigenvector with largest eigenvalue!
- And we're done with the $k = 1$ case!

By repeating a similar derivation for subsequent $\mathbf{w}_j$, we can find by induction that they are the subsequent eigenvectors.

Dealing with non-zero mean

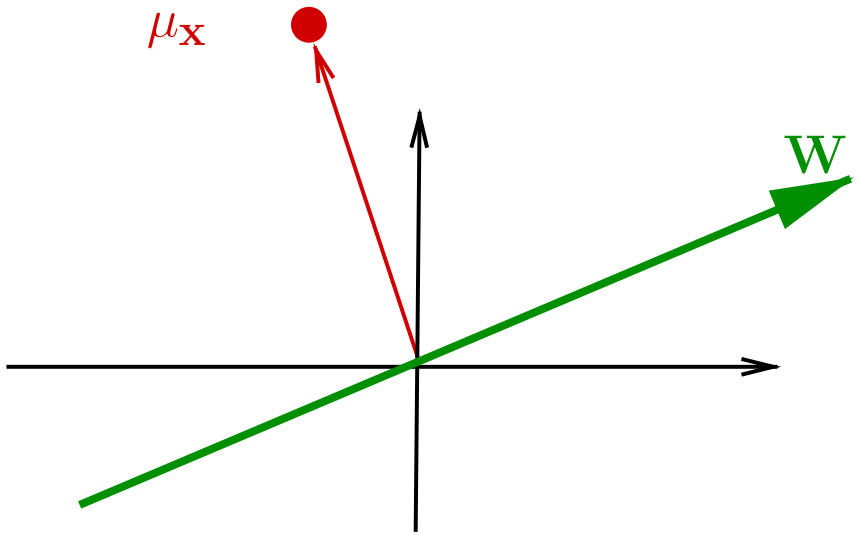
- Suppose the data mean is $\mu_x \neq \mathbf{0}$. Changes in solution:
- Covariance becomes

$$\frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \mu_{\mathbf{x}})(\mathbf{x}_i - \mu_{\mathbf{x}})^T$$



- Low-dimensional representation is given by

$$\mathbf{z} = \mathbf{W}^T(\mathbf{x} - \mu_{\mathbf{x}})$$



- Reconstruction:

$$\hat{\mathbf{x}} = \mu_{\mathbf{x}} + \mathbf{W}\mathbf{z}$$

Dealing with non-zero mean

- Suppose the data mean is $\mu_x \neq \mathbf{0}$. Changes in solution:
- Covariance becomes

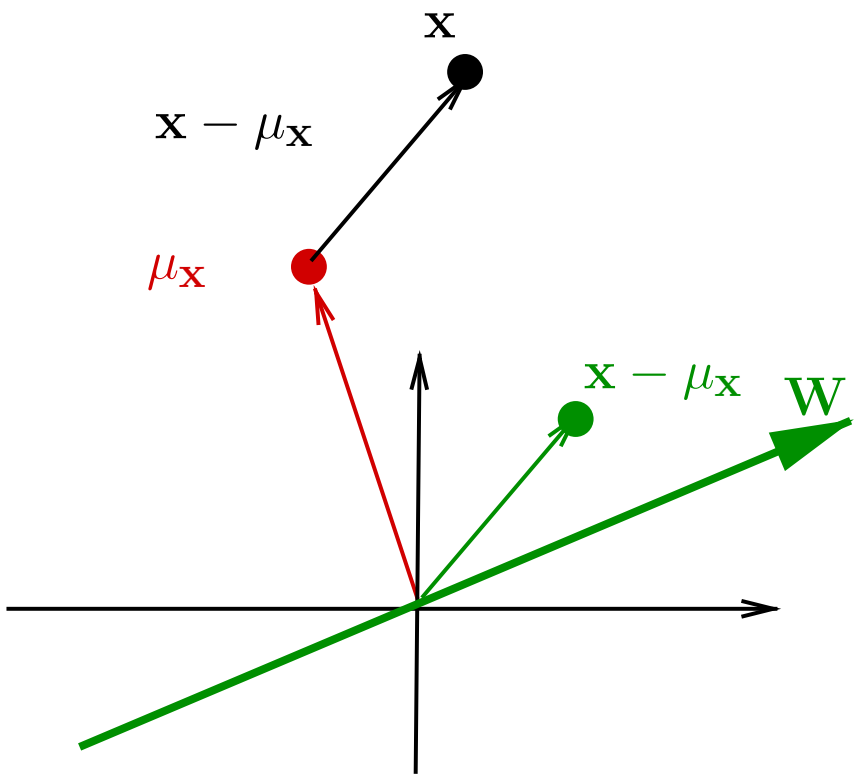
$$\frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i-\mu_{\mathbf{x}})(\mathbf{x}_i-\mu_{\mathbf{x}})^T$$

- Low-dimensional representation is given by

$$\mathbf{z} = \mathbf{W}^T(\mathbf{x}-\mu_{\mathbf{x}})$$

- Reconstruction:

$$\hat{\mathbf{x}} = \mu_{\mathbf{x}} + \mathbf{W}\mathbf{z}$$



Dealing with non-zero mean

- Suppose the data mean is $\mu_x \neq \mathbf{0}$. Changes in solution:
- Covariance becomes

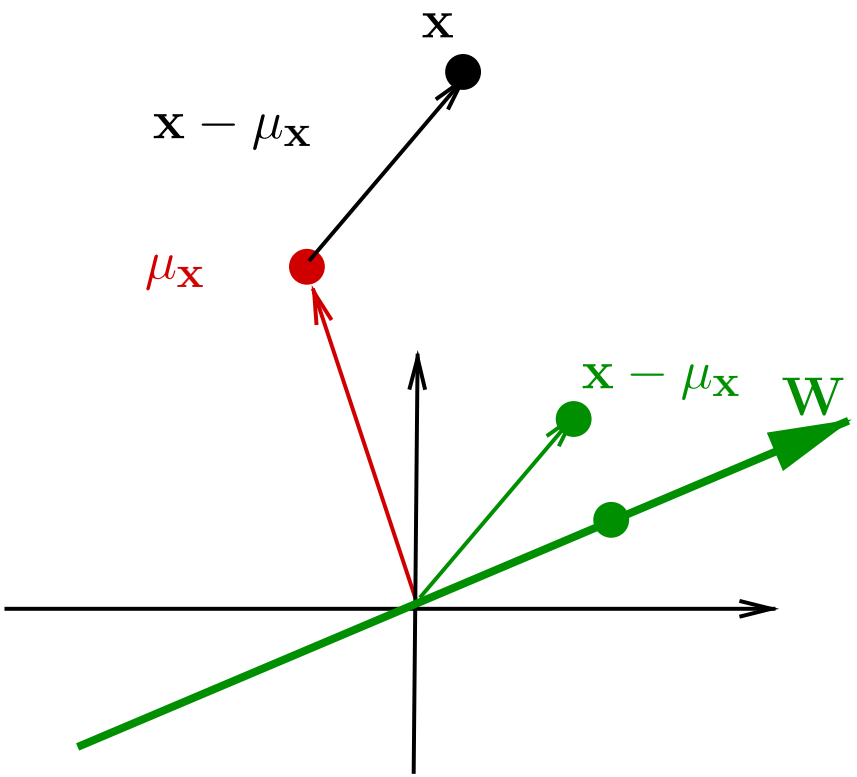
$$\frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i-\mu_{\mathbf{x}})(\mathbf{x}_i-\mu_{\mathbf{x}})^T$$

- Low-dimensional representation is given by

$$\mathbf{z} = \mathbf{W}^T(\mathbf{x}-\mu_{\mathbf{x}})$$

- Reconstruction:

$$\hat{\mathbf{x}} = \mu_{\mathbf{x}} + \mathbf{W}\mathbf{z}$$



Dealing with non-zero mean

- Suppose the data mean is $\mu_x \neq \mathbf{0}$. Changes in solution:
- Covariance becomes

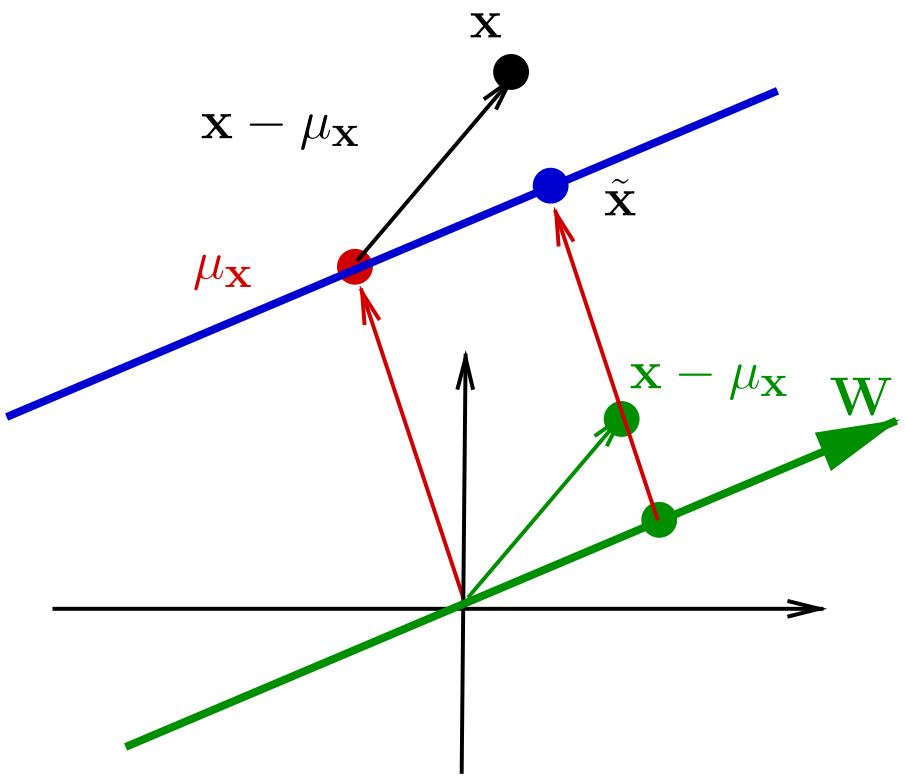
$$\frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \mu_{\mathbf{x}})(\mathbf{x}_i - \mu_{\mathbf{x}})^T$$

- Low-dimensional representation is given by

$$\mathbf{z} = \mathbf{W}^T(\mathbf{x} - \mu_{\mathbf{x}})$$

- Reconstruction:

$$\hat{\mathbf{x}} = \mu_{\mathbf{x}} + \mathbf{W}\mathbf{z}$$



Interpretation of PCA: Visualization in projected space

Interpretation of PCA: Visualization in projected space



Interpretation of PCA: Visualization of principal directions

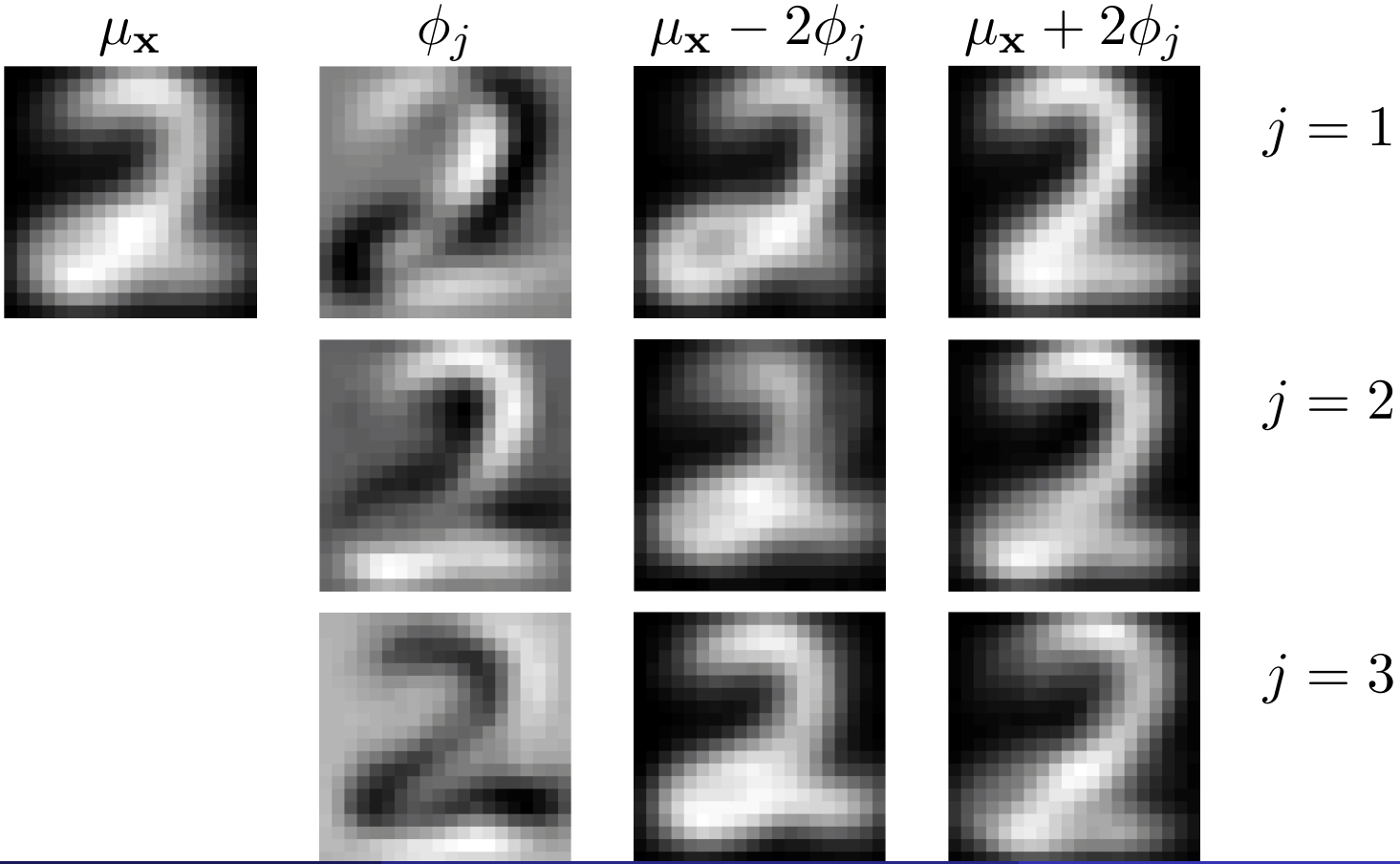
- Each principal direction $\phi_j = \mathbf{w}_j$ is a vector in the same space as $\mathbf{x}$; determines *direction*, not *location*.

Interpretation of PCA: Visualization of principal directions

- Each principal direction $\phi_j = \mathbf{w}_j$ is a vector in the same space as $\mathbf{x}$; determines *direction*, not *location*.
- Suppose each $\mathbf{x}$ is an image. How do we visualize ϕ_j?

Interpretation of PCA: Visualization of principal directions

- Each principal direction $\phi_j = \mathbf{w}_j$ is a vector in the same space as $\mathbf{x}$; determines *direction*, not *location*.
- Suppose each $\mathbf{x}$ is an image. How do we visualize ϕ_j?



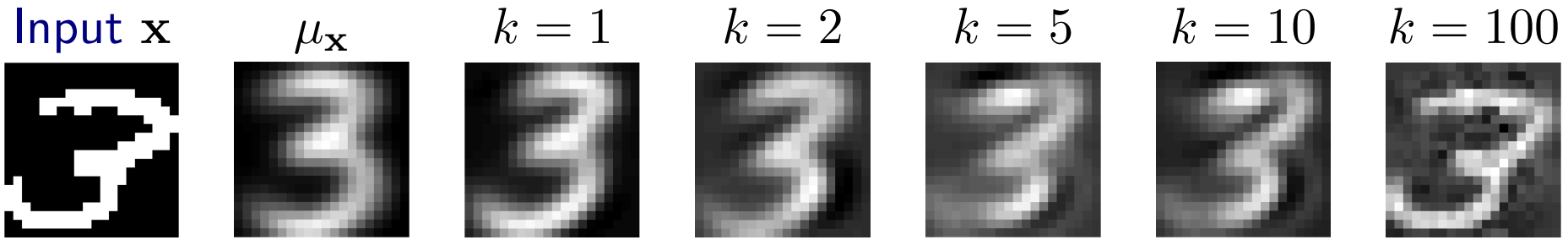
How should we choose the dimensionality k?

Choosing the dimensionality k

- The eigenvectors (columns of $\mathbf{\Phi}$) form a basis
- We can look at the expansion

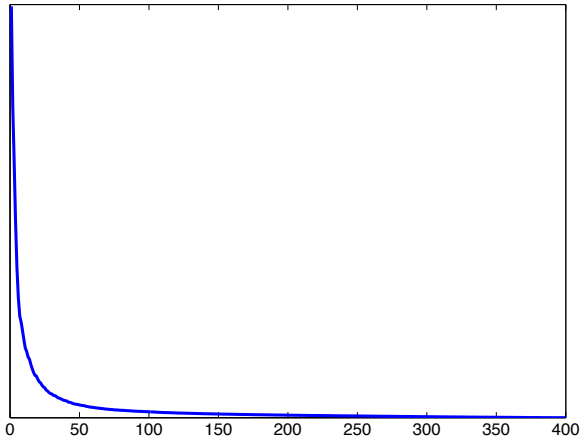
$$\hat{\mathbf{x}} = \mu_{\mathbf{x}} + \sum_{j=1}^{k} (\phi_j^T \mathbf{x}) \phi_j,$$

  and examine the residual $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$
- Expected residual $\Leftrightarrow$ variance in the remaining subspace.



Choosing the dimensionality k

- Typical plot of λs:



- Reasonable criteria:
  - Look for an "elbow" in the plot.
  - Set a threshold on percentage variance retained in the principal subspace:

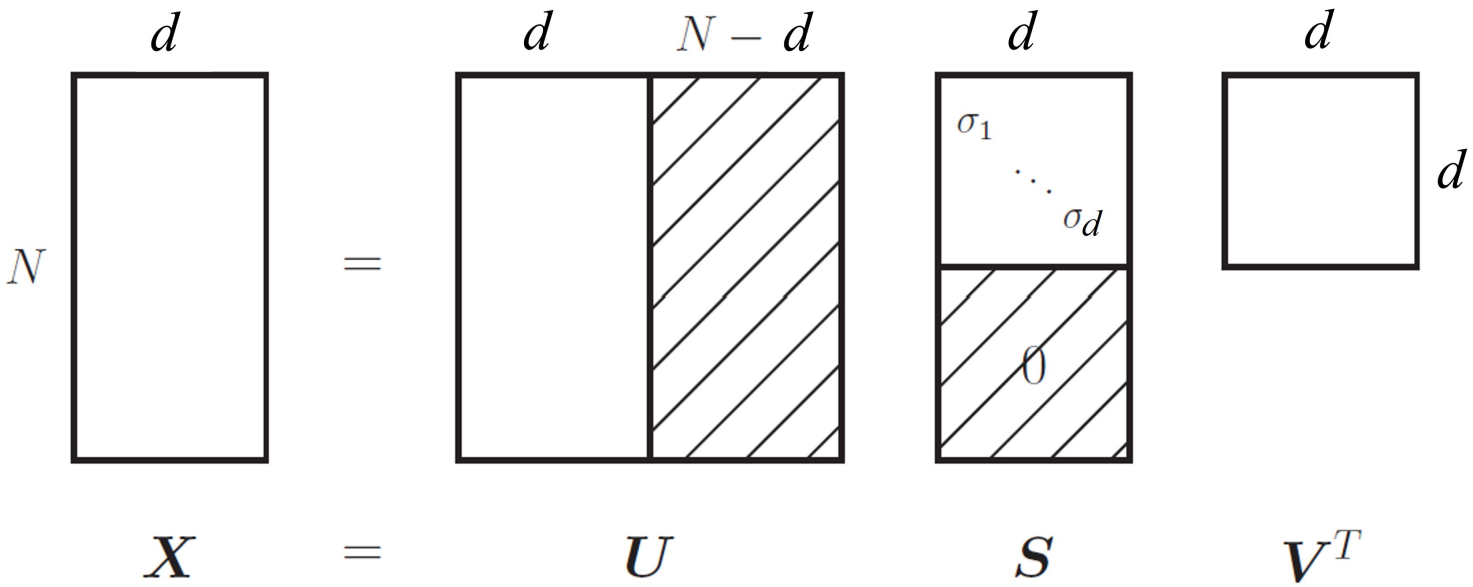
$$V(k) = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{l=1}^{d} \lambda_l}$$

  - Tune k for some downstream task performance

Relationship between PCA and singular value decomposition (SVD)

Recall: any real $N \times d$ matrix $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$

- $\mathbf{U} : N \times N$ where $\mathbf{U}^T\mathbf{U} = I$
- $\mathbf{S} : N \times d$ containing the $r = \min(N, d)$ singular values $\sigma_i \geq 0$ on the main diagonal and 0s elsewhere
- $\mathbf{V} : d \times d$ where $\mathbf{V}^T\mathbf{V} = \mathbf{V}\mathbf{V}^T = I$



Relationship between PCA and SVD (cont'd)

Recall: any real $N \times d$ matrix $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$

- $\mathbf{U} : N \times N$ where $\mathbf{U}^T\mathbf{U} = I$
- $\mathbf{S} : N \times d$ containing the $r = \min(N, d)$ singular values $\sigma_i \geq 0$ on the main diagonal and 0s elsewhere
- $\mathbf{V} : d \times d$ where $\mathbf{V}^T\mathbf{V} = \mathbf{V}\mathbf{V}^T = I$

Properties:

- The right singular vectors $\mathbf{V}$ are equal to the eigenvectors of $\mathbf{X}^T\mathbf{X}$
- The eigenvalues of $\mathbf{X}^T\mathbf{X}$ are the squared singular values σ_i

So, if $\mathbf{X}$ is a data matrix, and $\mathbf{X}^T\mathbf{X}$ is its empirical covariance, we can compute the PCA solution via an SVD

PCA and compression

- Suppose we have computed k-dimensional PCA representation.
- We need to transmit/store:
  - The $1 \times d$ mean vector;
  - The $k \times d$ projection matrix.
- For each new example, we only need to convey $\mathbf{z}$ which is $1 \times k$.
  - If we transmit N examples, we need $d + dk + Nk$ numbers instead of Nd.
  - Tradeoff between accuracy and compression
  - (We'll explore this a bit in HW1)

PCA and classification

- A very common approach: perform PCA on all data and learn a classifier in the low-dimensional space
- Can greatly speed up classifier training

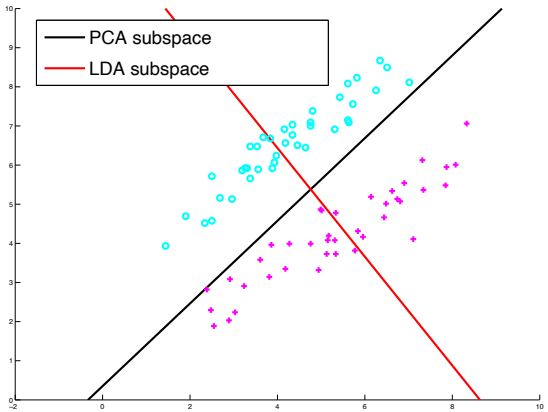
PCA and classification

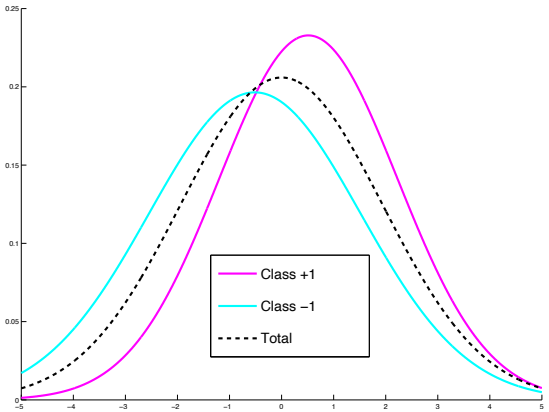
- A very common approach: perform PCA on all data and learn a classifier in the low-dimensional space
- Can greatly speed up classifier training
- Alternative setting: perform PCA on large amount of unlabeled data, learn classifier on small amount of labeled data after dimensionality reduction
- Might learn *more accurate* classifier than by learning on the raw labeled data set
- (This will also be explored in HW1)

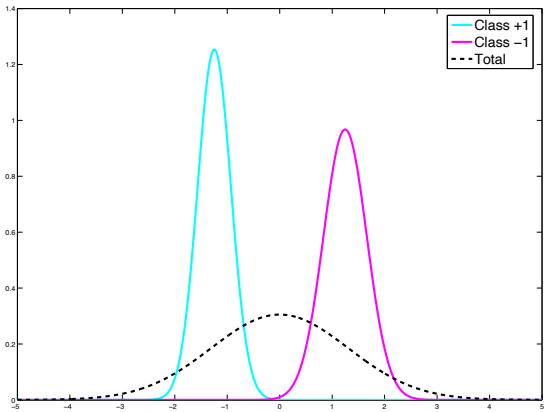
PCA and classification (continued)

However, PCA can be "fooled": Direction of largest variance need not be the most *discriminative* direction

- If we knew the labels, we could use a supervised dimensionality reduction (e.g., linear discriminant analysis)
- Consider the original data, projection along top PCA direction, and projection along most discriminative direction (from LDA):







Multi-view linear dimensionality reduction

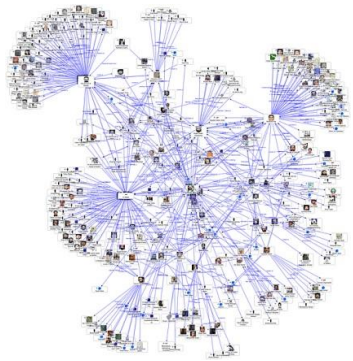
Multi-view linear dimensionality reduction

- Unsupervised dimensionality reduction is very challenging
- If we happen to have multiple data "views", it can be easier

Multi-view data examples



+



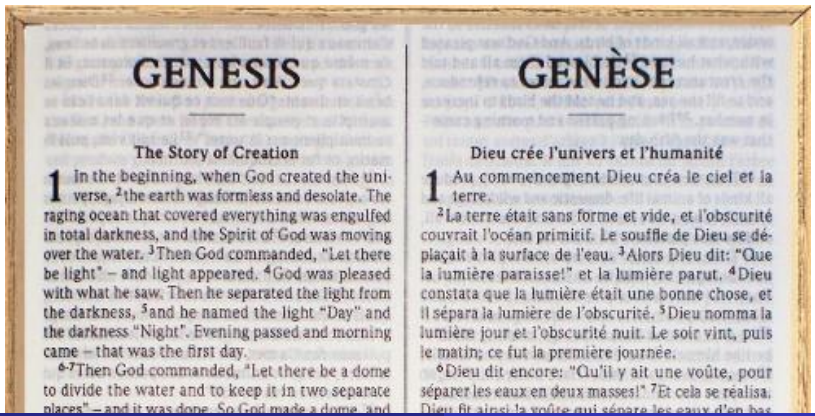


Family with 2 kids and a dog

+

Several kinds of winter squash

The Earth as seen from space



GENESIS

The Story of Creation

1 In the beginning, when God created the universe, 2 the earth was formless and desolate. The raging ocean that covered everything was engulfed in total darkness, and the Spirit of God was moving over the water. 3 Then God commanded, "Let there be light" – and light appeared. 4 God was pleased with what he saw. Then he separated the light from the darkness, 5 and he named the light "Day" and the darkness "Night". Evening passed and morning came – that was the first day.

6-7 Then God commanded, "Let there be a dome to divide the water and to keep it in two separate

GENÈSE

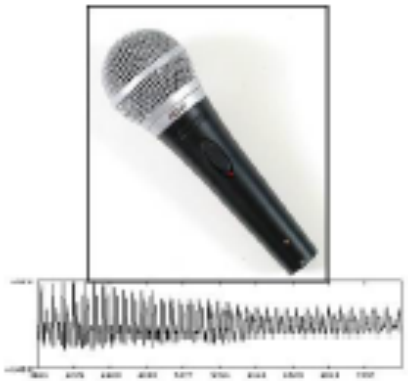
Dieu crée l'univers et l'humanité

1 Au commencement Dieu créa le ciel et la terre.

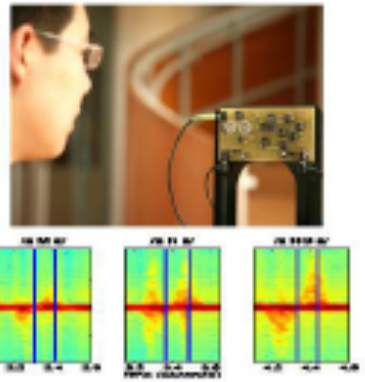
2 La terre était sans forme et vide, et l'obscurité couvrait l'océan primitif. Le souffle de Dieu se déplaçait à la surface de l'eau. 3 Alors Dieu dit: "Que la lumière paraisse!" et la lumière parut. 4 Dieu constata que la lumière était une bonne chose, et il sépara la lumière de l'obscurité. 5 Dieu nomma la lumière jour et l'obscurité nuit. Le soir vint, puis le matin; ce fut la première journée.

6 Dieu dit encore: "Qu'il y ait une voûte, pour séparer les eaux en deux masses!" 7 Et cela se réalisa.

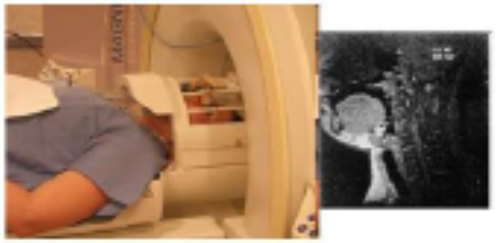
Multi-view data examples: Speech









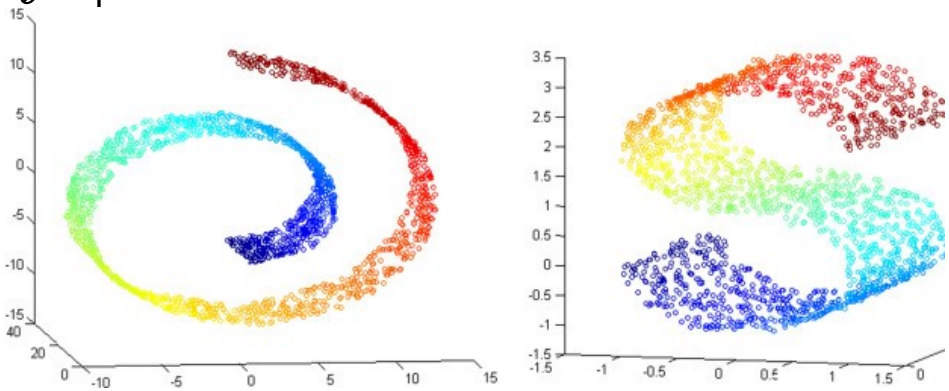




Multi-view representation learning (linear or not)

Training data consists of samples of a d-dimensional random vector that has some natural split into two sub-vectors:

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \ \mathbf{x} \in \mathbf{R}^{d_x}, \ \mathbf{y} \in \mathbf{R}^{d_y}, \ d_x + d_y = d$$



- **Multi-view representation learning**: The problem of learning useful features/subspaces from such two-view data
- Typically involves learning representations of one view that are **predictive** of the other view, and vice versa
- Intuition: If the noise/nuisance parameters in the two views are independent, then the common information must be signal!

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$
- First pair of directions:

$$v_1, w_1 \quad = \quad \underset{v,w}{\operatorname{argmax}} \ \operatorname{corr}(v^T X, w^T Y)$$

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$
- First pair of directions:

$$\begin{aligned} v_1, w_1 &= \underset{v,w}{\operatorname{argmax}} \ \operatorname{corr}(v^T X, w^T Y) \\ &= \underset{v,w}{\operatorname{argmax}} \ \frac{v^T C_{xy} w}{\sqrt{(v^T C_{xx} v)(w^T C_{yy} w)}} \end{aligned}$$

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$
- First pair of directions:

$$\begin{aligned} v_1, w_1 &= \underset{v,w}{\operatorname{argmax}} \ \operatorname{corr}(v^T X, w^T Y) \\ &= \underset{v,w}{\operatorname{argmax}} \ \frac{v^T C_{xy} w}{\sqrt{(v^T C_{xx} v)(w^T C_{yy} w)}} \end{aligned}$$

- Subsequent direction vectors maximize the same correlation, subject to being uncorrelated with the previous directions

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$
- First pair of directions:

$$\begin{aligned} v_1, w_1 &= \underset{v,w}{\operatorname{argmax}} \ \operatorname{corr}(v^T X, w^T Y) \\ &= \underset{v,w}{\operatorname{argmax}} \ \frac{v^T C_{xy} w}{\sqrt{(v^T C_{xx} v)(w^T C_{yy} w)}} \end{aligned}$$

- Subsequent direction vectors maximize the same correlation, subject to being uncorrelated with the previous directions
- Equivalent to maximizing the covariance, subject to $v^T C_{xx} v = 1, w^T C_{yy} w = 1$

Canonical correlation analysis (CCA)

Solution can be found via an eigenproblem:

$$\begin{aligned} C_{xx}^{-1} C_{xy} C_{yy}^{-1} C_{yx} v &= \lambda^2 v \\ w &\propto C_{yy}^{-1} C_{yx} v \end{aligned}$$

Canonical correlation analysis (CCA)

Solution can be found via an eigenproblem:

$$\begin{aligned} C_{xx}^{-1} C_{xy} C_{yy}^{-1} C_{yx} v &= \lambda^2 v \\ w &\propto C_{yy}^{-1} C_{yx} v \end{aligned}$$

Canonical correlation analysis (CCA)

Solution can be found via an eigenproblem:

$$\begin{aligned} C_{xx}^{-1} C_{xy} C_{yy}^{-1} C_{yx} v &= \lambda^2 v \\ w &\propto C_{yy}^{-1} C_{yx} v \end{aligned}$$

Can use CCA for dimensionality reduction of just one view, say View 1 (X):

- Compute first k CCA directions
- Project new data from View 1 onto those directions

Canonical correlation analysis (CCA)

Alternative formulations: Find $V \in \mathbf{R}^{d_x \times k}, W \in \mathbf{R}^{d_y \times k}$ that

$$(1) \qquad \text{minimize}: \quad E||W^T Y - V^T X||_F^2$$

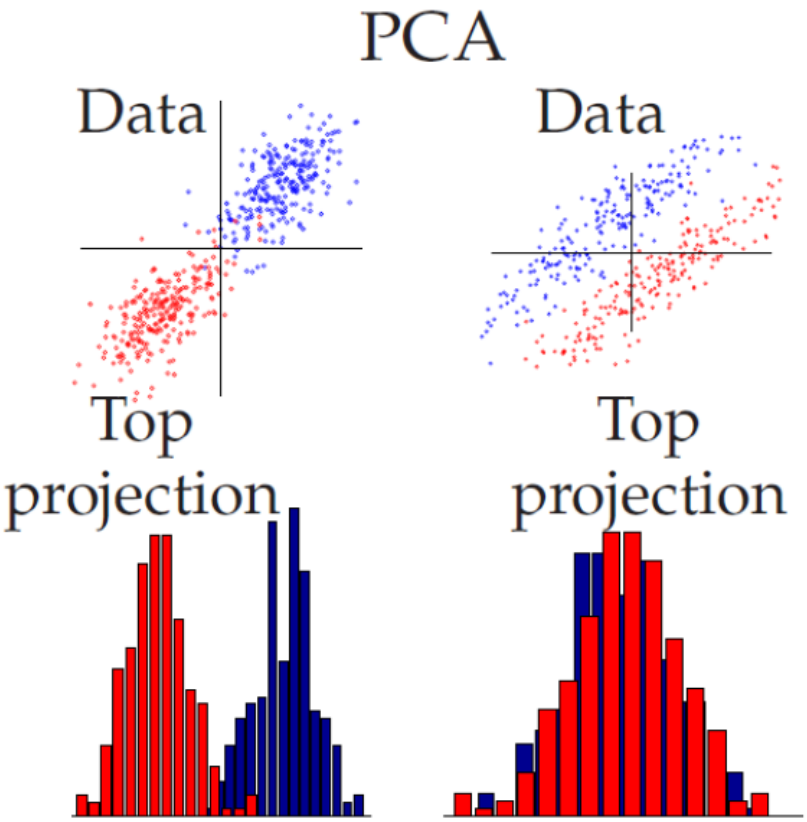
$$\text{subject to}: \quad V^T C_{xx} V = W^T C_{yy} W = I$$

Canonical correlation analysis (CCA)

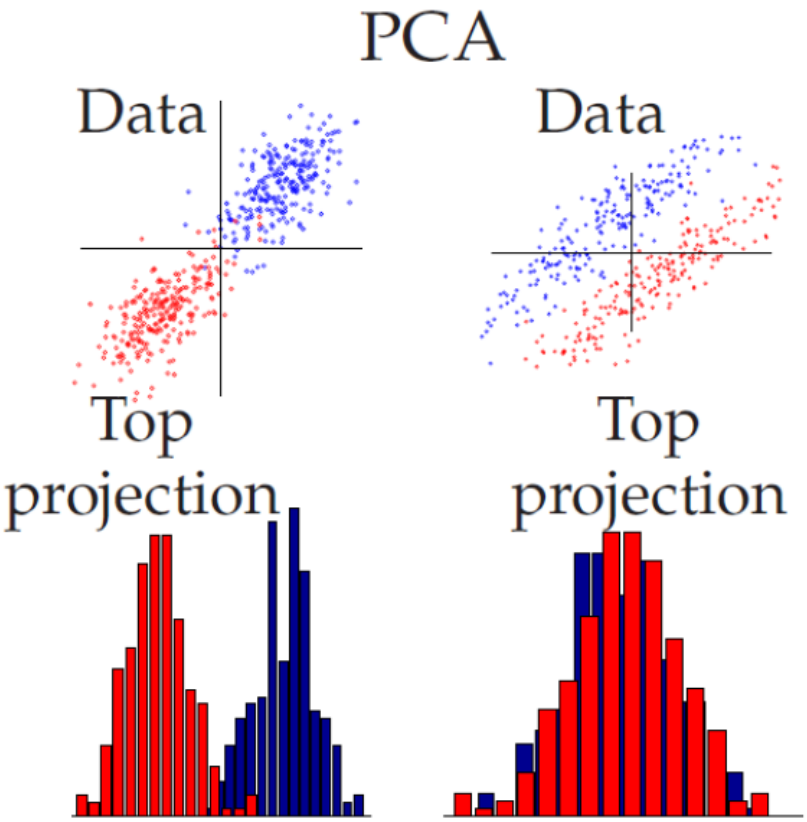
Alternative formulations: Find $V \in \mathbf{R}^{d_x \times k}, W \in \mathbf{R}^{d_y \times k}$ that

$$\begin{aligned} & (1) && \text{minimize}: && E||W^T Y - V^T X||_F^2 \\ OR\ & (2) && \text{maximize}: && \operatorname{Tr} V^T C_{xy} W \\ & && \text{subject to}: && V^T C_{xx} V = W^T C_{yy} W = I \end{aligned}$$

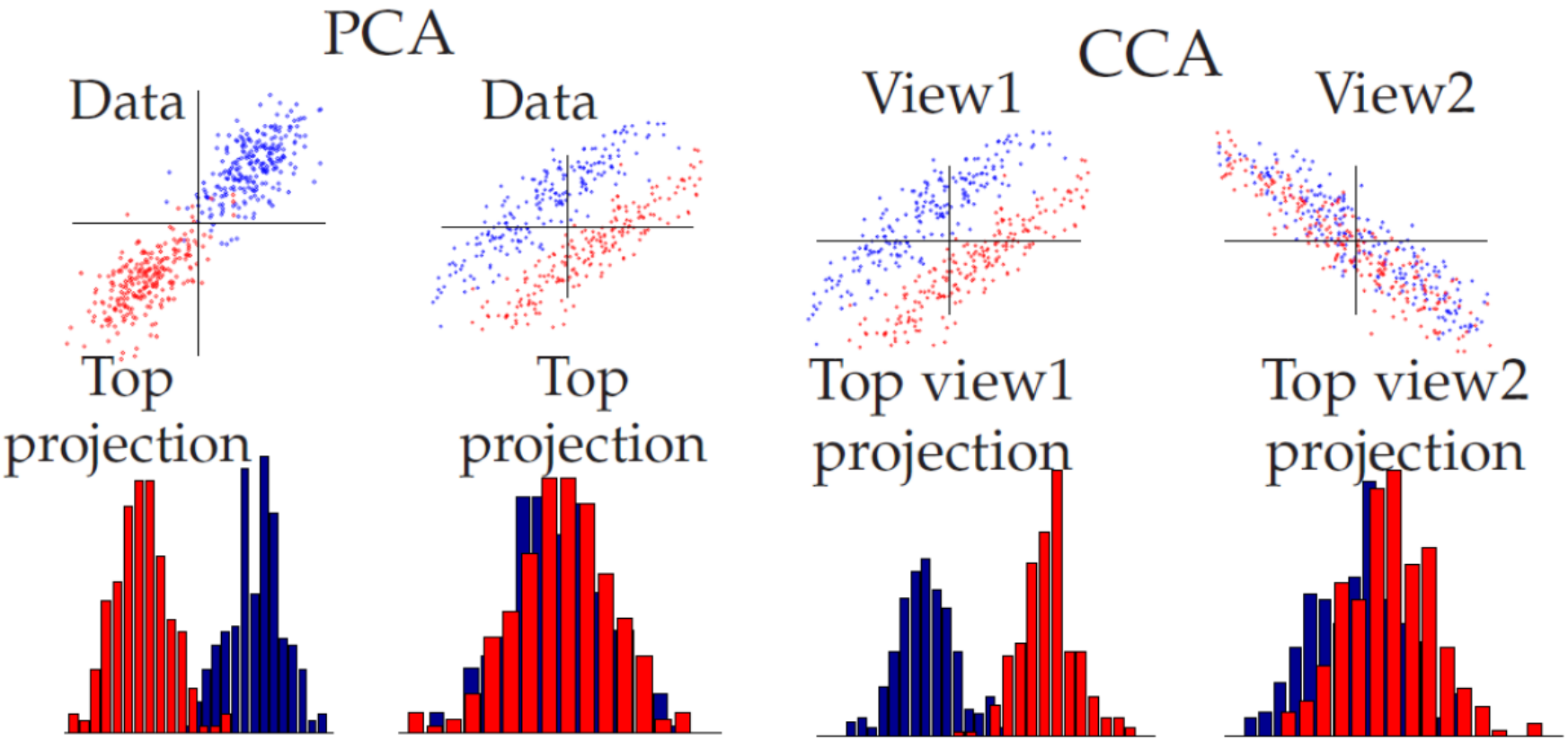
Toy examples



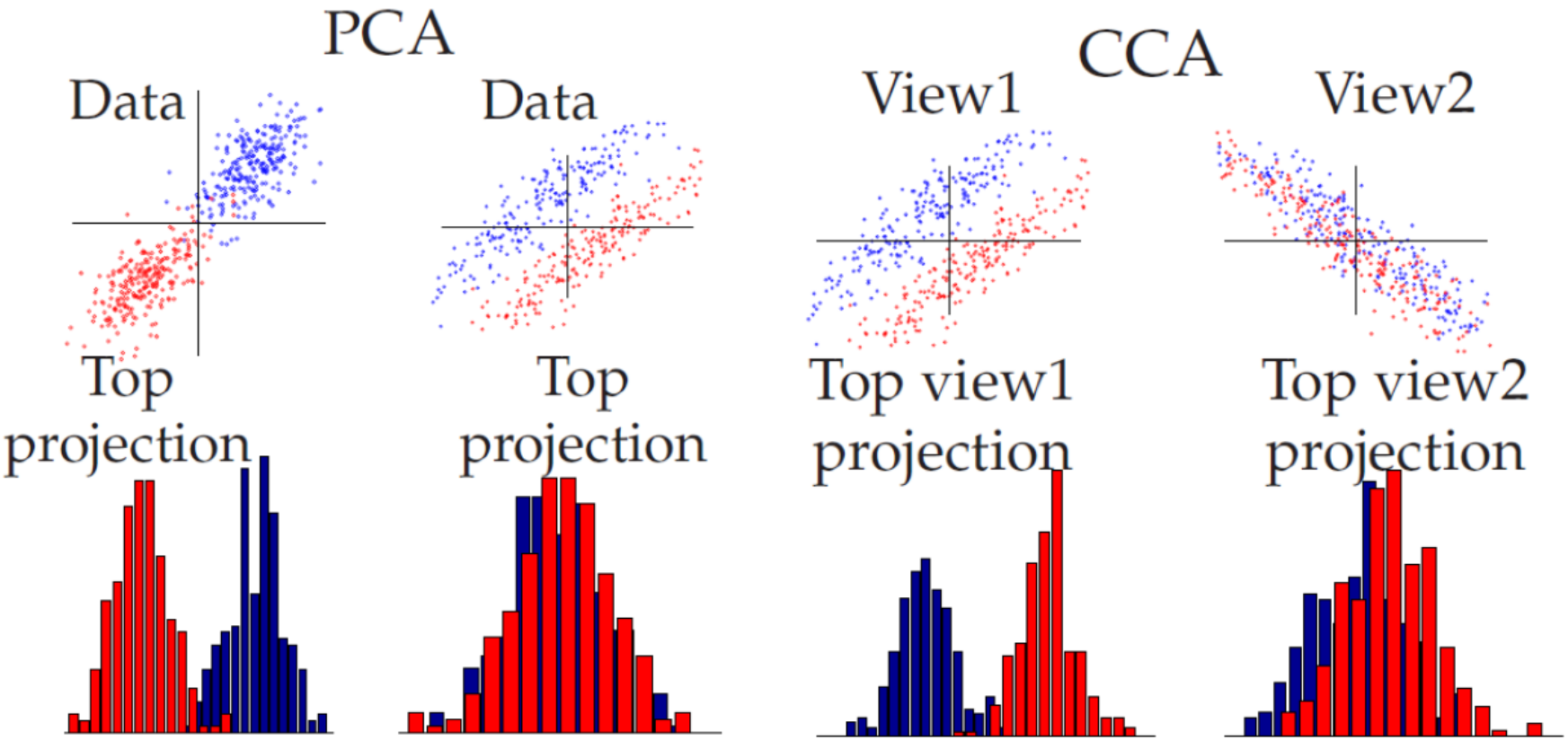
Toy examples



Toy examples



Toy examples



- If there is a class structure in the data, CCA has discriminative properties when the two views are uncorrelated given the class label

Regularizing CCA

CCA can overfit, i.e. to find spurious correlations in the training data

Regularizing CCA

CCA can overfit, i.e. to find spurious correlations in the training data

Solution: Regularize. Instead of original problem, solve

$$v_1, w_1 = \underset{v,w}{\operatorname{argmax}} \frac{v^T C_{xy} w}{\sqrt{v^T (C_{xx} + r_x I) v w^T (C_{yy} + r_y I) w}}$$

Regularizing CCA

CCA can overfit, i.e. to find spurious correlations in the training data

Solution: Regularize. Instead of original problem, solve

$$v_1, w_1 = \underset{v,w}{\operatorname{argmax}} \frac{v^T C_{xy} w}{\sqrt{v^T (C_{xx} + r_x I) v w^T (C_{yy} + r_y I) w}}$$

Solution becomes eigenvectors of $(C_{xx} + r_x I)^{-1} C_{xy} (C_{yy} + r_y I)^{-1} C_{yx}$

Optionally, also do an initial dimensionality reduction via PCA in each view

Multi-view dimensionality reduction applications

All the usual applications of dimensionality reduction: compression, improved classification with limited labeled data, ... but also:

- Cross-view retrieval
- To "line up" two views for synchronization/visualization/analysis

Example recent application domains: sound source localization in images, image-text retrieval, cross-lingual word embeddings, speech recognition, neuroscience, ...